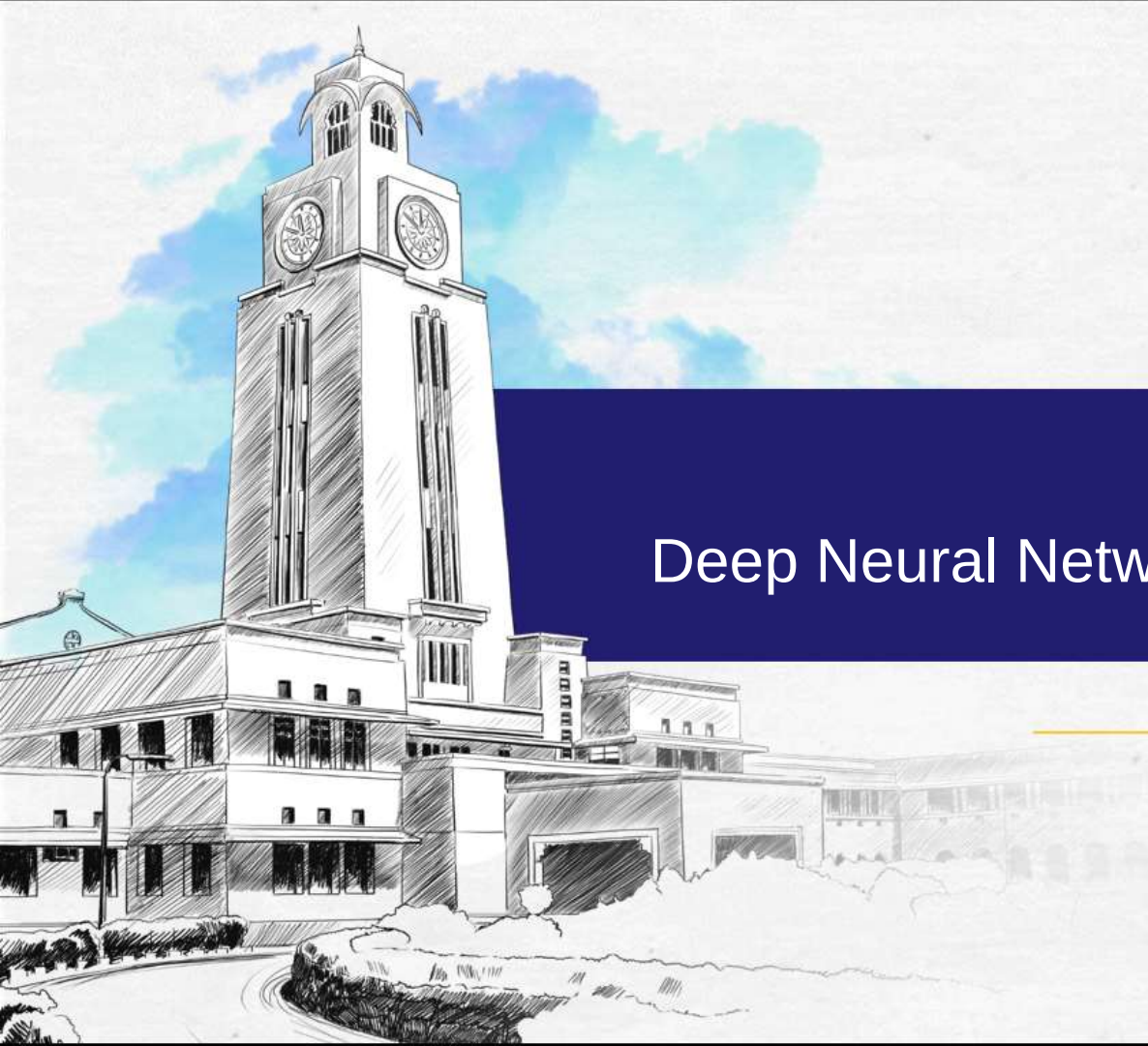




BITS Pilani
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Deep Neural Network(AIML ZG565)

Prof. Rajiv Ranjan Gupta

BITS Pilani

Course Logistics



What we Learn.... (Module Structure)

1. Deep learning - What, Why, where
2. Artificial Neuron and Perceptron
3. Linear Neural Network for Regression
4. Linear Neural Network for Classification
5. Deep Feedforward Neural Network
6. Convolutional Neural Network
7. Recurrent Neural Networks
8. Attention Mechanism
9. Transformer
10. Optimization of Deep models
11. Regularization for Deep models

Books

Text book

- Dive into Deep Learning by Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola. https://d2l.ai/chapter_introduction/index.html

Reference book

- Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville
<https://www.deeplearningbook.org/>

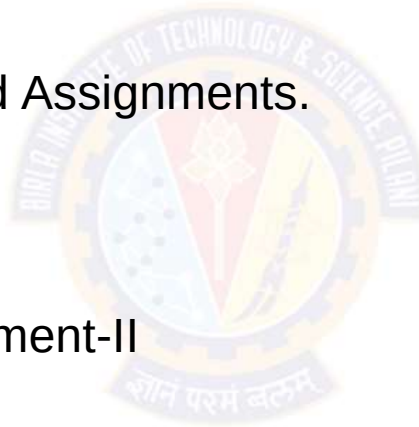
Evaluation Components and Schedule

Component	Name	Weightage	Type
EC 1	Quiz I	5 %	Pre-midsem
	Quiz II	5 %	Post-midsem
	Practice Lab Assignment I	10 %	Pre-midsem
	Situated Learning Assignment II	10 %	Post-midsem
EC 2	Mid sem Exam	30 %	Sessions 1-8
EC 3	Compre Exam	40 %	Sessions 1 -

EC1 = Quiz I + Quiz II + Assignment I + Assignment II

Course Logistics

- Refer Taxila for the following
 - Handout
 - Schedule for Webinar
 - Schedule of Quiz, and Assignments.
 - Session Slide Deck
 - Demo Lab Sheets
 - Quiz-I, Quiz-II
 - Assignment-I, Assignment-II
 - Sample QPs
- Virtual Lab
 - Practice lab sheets
- Lecture Recordings
 - Available on Microsoft teams



Honour Code

All submissions for graded components must be the result of your original effort. It is strictly prohibited to copy and paste verbatim from any sources, whether online or from your peers. The use of unauthorized sources or materials, as well as collusion or unauthorized collaboration to gain an unfair advantage, is also strictly prohibited. Please note that we will not distinguish between the person sharing their resources and the one receiving them for plagiarism, and the consequences will apply to both parties equally.

In cases where suspicious circumstances arise, such as identical verbatim answers or a significant overlap of unreasonable similarities in a set of submissions, will be investigated, and severe punishments will be imposed on all those found guilty of plagiarism.

In case of Queries regarding the course....

Step 1: Post in the discussion forum.

- Read through the existing post and if you find any topic similar to your concern, add on to the existing discussion.
- Avoid duplication of queries or issues.

Step 2: Email to the IC at seetha.p@pilani.bits-pilani.ac.in if the query or issue is not resolved within 1 weeks time. Turn around for response to the email is 48hrs.

- In the subject pl mention the phrase "DNN" clearly.
- **Use BITS email id for correspondence.** Emails from personal emails will be ignored without any reply.

PATIENCE is highly APPRECIATED :)

What is Deep Learning?

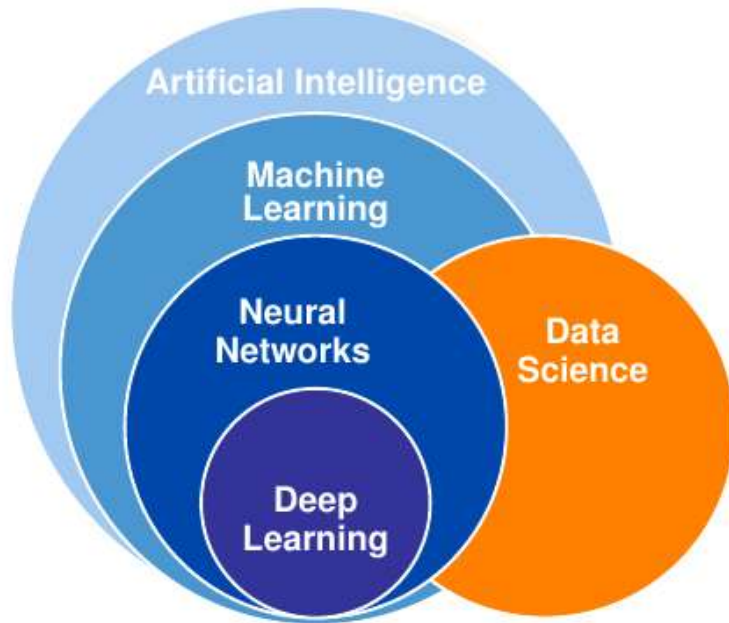


Definitions of Deep Learning

- Deep Learning is a type of **machine learning** based on **artificial neural networks** in which multiple layers of processing are used to extract progressively higher level features from data.
- Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is **inspired by the human brain**.
- Deep learning is a machine learning technique that teaches computers to do what comes naturally to humans: **learn by example**.
- Deep learning is a subset of machine learning, which is essentially a neural network with **three or more layers**.
- Deep Learning gets its name from the fact that we add more "**Layers**" to learn from the data.

Where in AI sits DL?

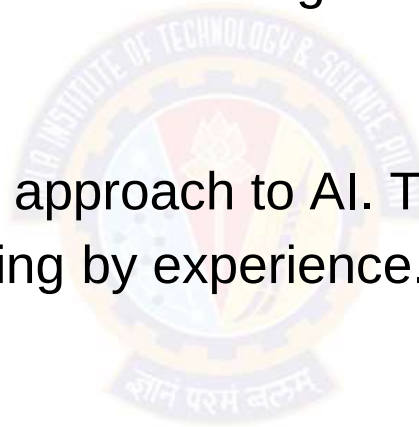
AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that don't involve any learning.



AI : Artificial intelligence is the science of making things smart. The aim is make machines perform human tasks. Eg: Robot cleaning a room.

ML : Machine learning is an approach to AI. The machine learns or perform tasks through learning by experience.

DL : Deep Learning is a technique for implementing machine learning to recognise patterns.



Deep (Machine) Learning

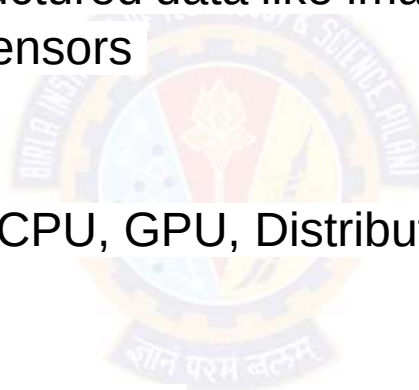
- Deep learning is a specific subfield of machine learning.
- Learning representations from data that puts an emphasis on learning successive layers of increasingly meaningful representations.
- The **deep** in deep learning stands for this idea of successive layers of representations.
- The number of layers that contribute to model the data is called the **depth** of the model.
- In deep learning, the layered representations are learned via models called **neural networks**, structured in literal layers stacked on top of each other.

Why Deep Learning?



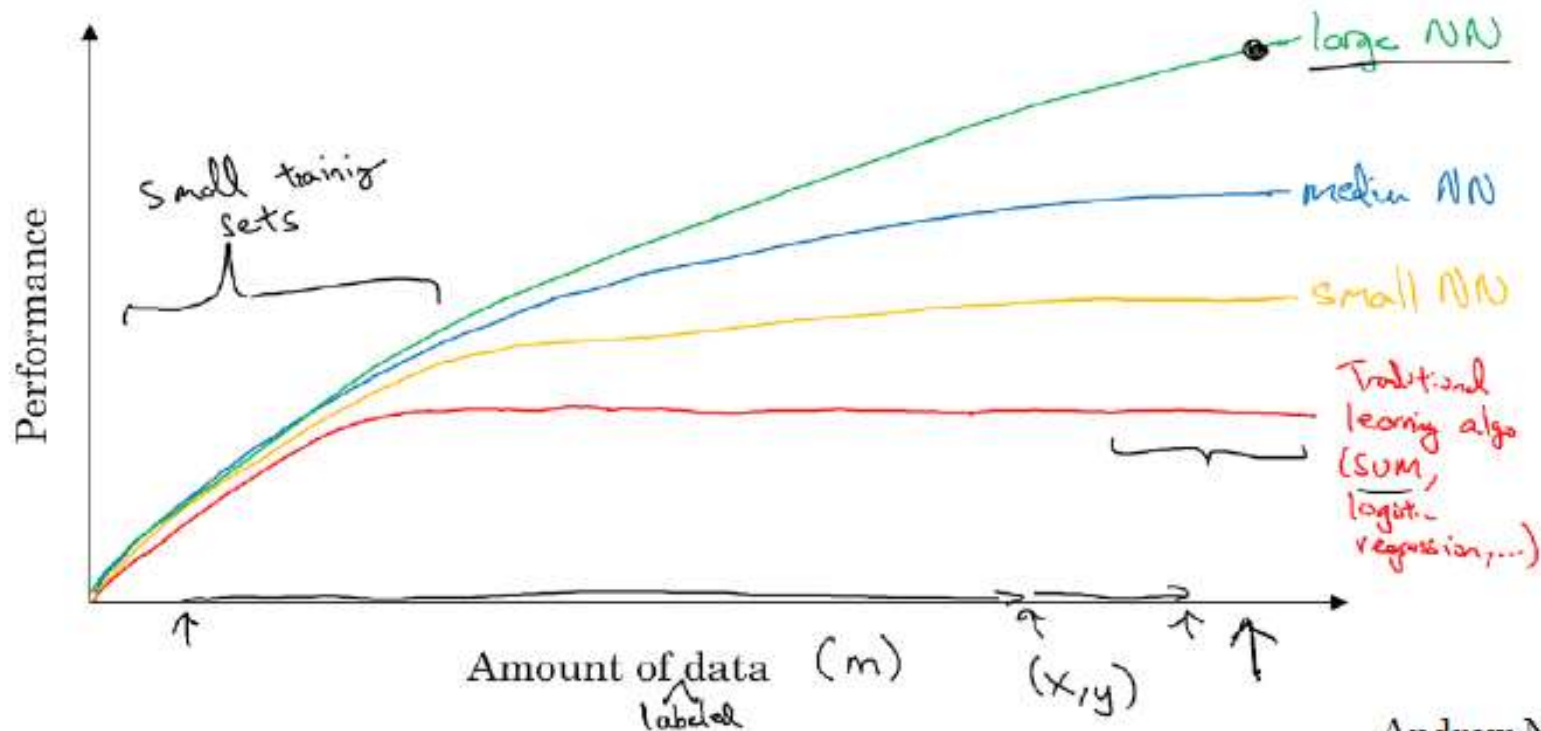
Why Deep Learning?

- Data and Storage
 - Large amounts of data
 - Lots and lots of unstructured data like images, text, audio, video
 - Cheap, high-quality sensors
 - Cheap data storage
- Computation
 - Cheap computation - CPU, GPU, Distributed clusters
- Algorithms
 - Learn by examples
 - Automated feature generation
 - Better learning capabilities
 - Scalability
 - Advance analytics can be applied

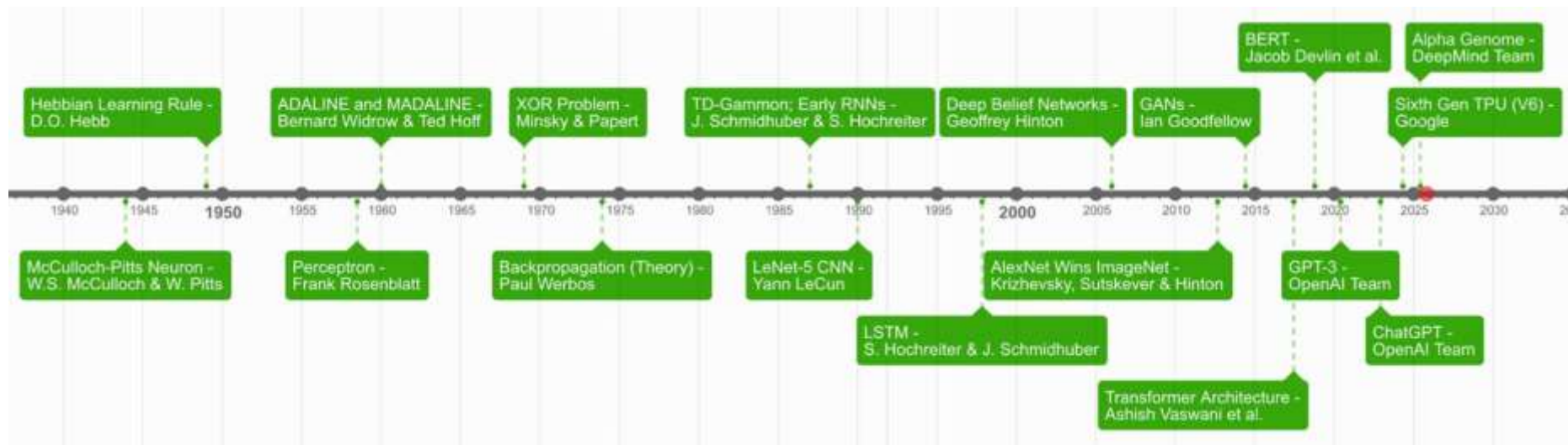


Why deep learning?

Scale drives deep learning progress



Deep Learning Timeline



Applications of Deep Learning



Breakthroughs with Neural Networks

TECHNEWSWORLD**EMERGING TECH**

SEARCH

ComputingInternetITMobile TechReviewsSecurityTechnologyTech BlogReader Service

Reasoning

Science

Knowledge

Learning

Computer

Technology

Artificial Intelligence

AI

Print

Email

Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

Most Popular

Newsletters

News Alert

How do you feel about Black Friday and Cyber Monday?

- They're great -- I get a lot of bargains!
- The deals are too spread out -- I'd prefer just one day.
- They're a fun way to kick off the holiday season.
- I don't like the commercialization of Thanksgiving Day.
- They're crucial for the retail industry and the economy.
- The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

AWS Enjoys Top Perch in IaaS, PaaS Markets

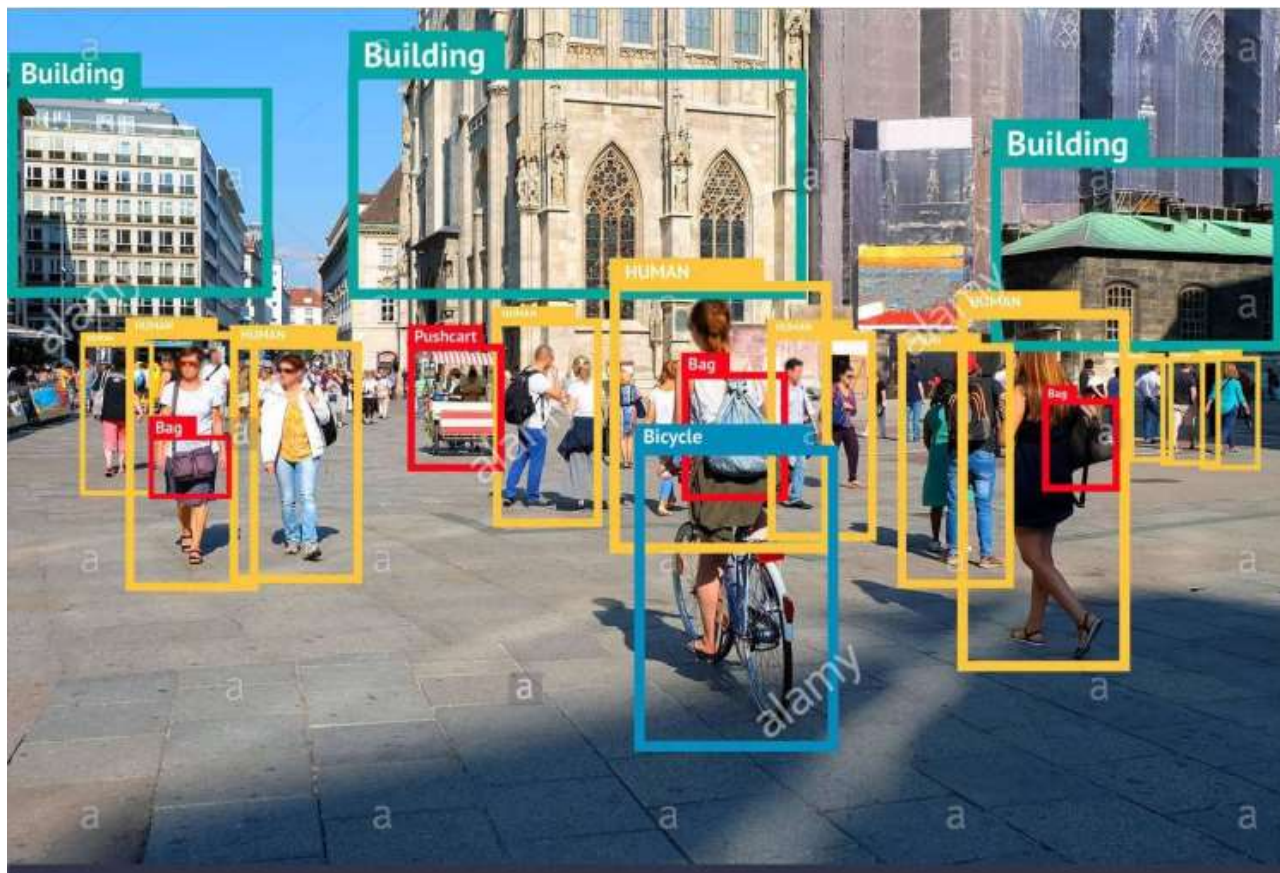
US Comptroller Gears Up for Blockchain and

19

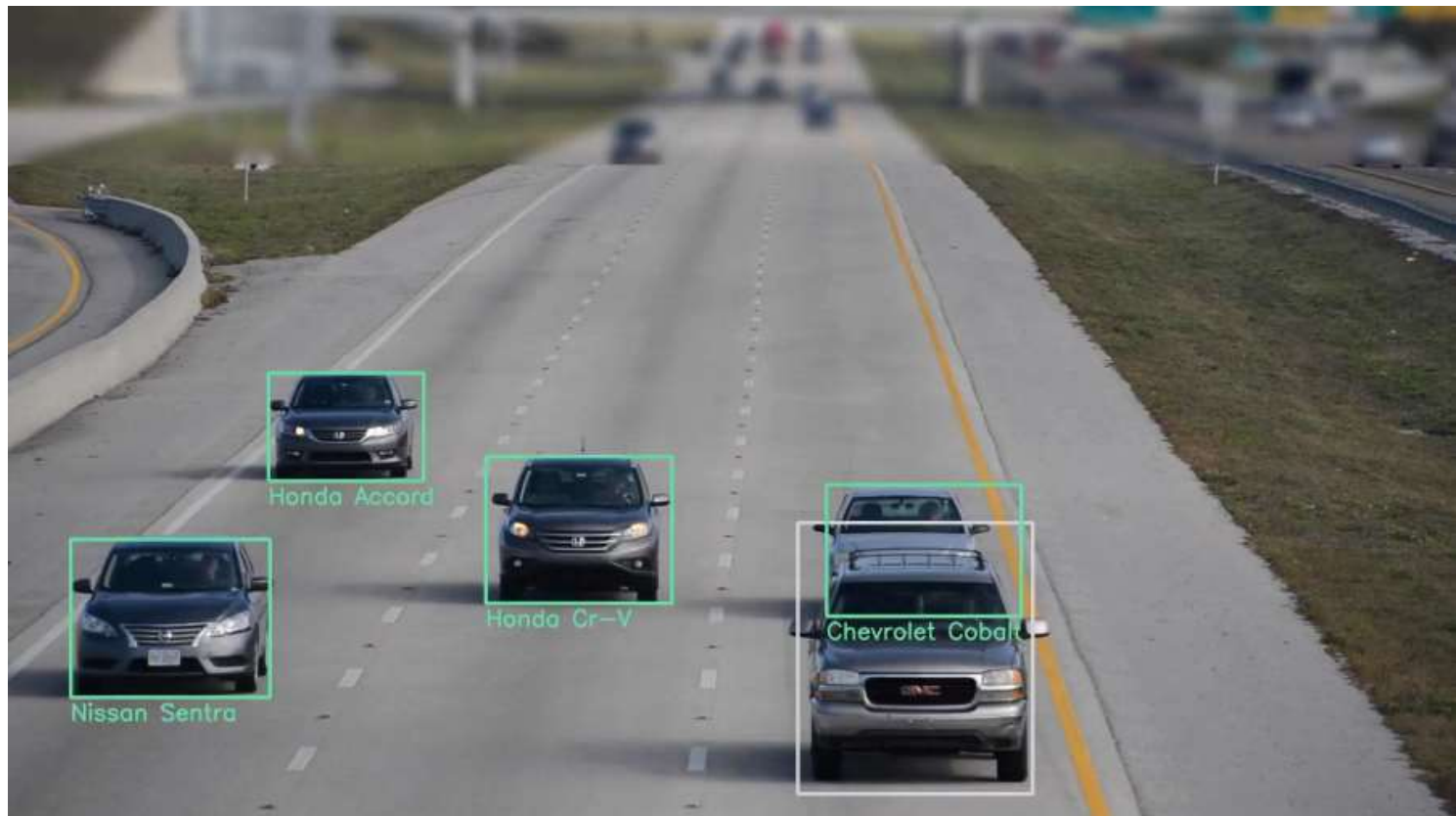
Breakthroughs with Neural Networks



Breakthroughs with Neural Networks



Breakthroughs with Neural Networks



Breakthroughs with Neural Networks

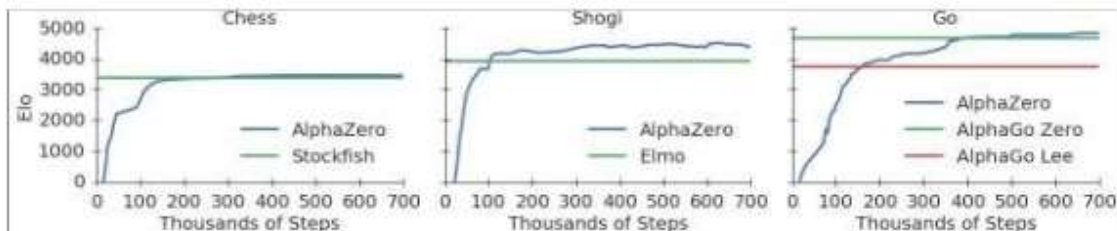


Figure 1: Training *AlphaZero* for 700,000 steps. Elo ratings were computed from evaluation games between different players when given one second per move. **a** Performance of *AlphaZero* in chess, compared to 2016 TCEC world-champion program *Stockfish*. **b** Performance of *AlphaZero* in shogi, compared to 2017 CSA world-champion program *Elmo*. **c** Performance of *AlphaZero* in Go, compared to *AlphaGo Lee* and *AlphaGo Zero* (20 block / 3 day) (29).

Breakthroughs with Neural Networks

Introducing ChatGPT

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.

[Try ChatGPT ↗](#)

[Read about ChatGPT Plus](#)

Applications of Deep Learning

Application	Input	Output	Neural Network
Real Estate	House features	House Price	Std NN
Photo Tagging	Image	Text	CNN
Object detection	Image	Bounding box	CNN
Speech Recognition	Audio	Text transcript	RNN
Translation	English Text	French Text	RNN
Autonomous driving	Image, Sensors Radars	Position of other cars, objects, signals	Hybrid NN

Many more applications....

- Predict tomorrow's weather given geographic information, satellite images, and a trailing window of past weather.
- Takes in a question, expressed in free-form text, and answers it correctly. (Q&A)
- Takes in an image and identify all the people it contains, drawing outlines around each. (Object detection)
- Presents users with products that they are likely to enjoy but unlikely, in the natural course of browsing, to encounter. (Recommendation systems)

Key components of DL problem



Core components of DL problem

1. The **data** that we can learn from.
2. A **model** of how to transform the data.
3. An **objective function** that quantifies how well (or badly) the model is doing.
4. An **algorithm** to adjust the model's parameters to optimize the obje



1. Data

- Collection of examples.
- Data has to be converted to an useful and a suitable numerical **representation**.
- Each example (or data point, data instance, sample) typically consists of a set of attributes called **features** (or covariates), from which the model must make its predictions.
- In the supervised learning problems, the attribute to be predicted is designated as the **label** (or target).

$$Data = \mathcal{D} = \{X, t\}$$

1. Data

- Data = $D = \{X, t\}$
- Number of features = Number of Dimensions = d
- Number of examples = N
- As the number of features increase, we say the dimensionality increases.



Examples

Features

(Target)

ID	Bedrooms	Location	Sq Ft	Price
1	3	Urban	1500	350K
2	2	Suburban	1200	280K
3	4	Urban	2000	520K
4	3	Rural	1800	310K
5	2	Suburban	1100	245K
6	5	Urban	2500	680K
7	3	Rural	1600	275K
8	4	Suburban	1900	425K

2. Model

- Model denotes the **computational machinery** for ingesting data of one type, and spitting out predictions of a possibly different type.
- Deep learning models consist of many successive transformations of the data that are chained together top to bottom, thus the name deep learning.

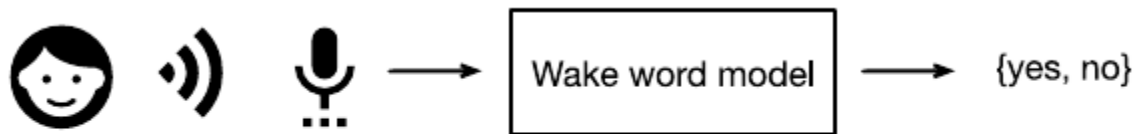
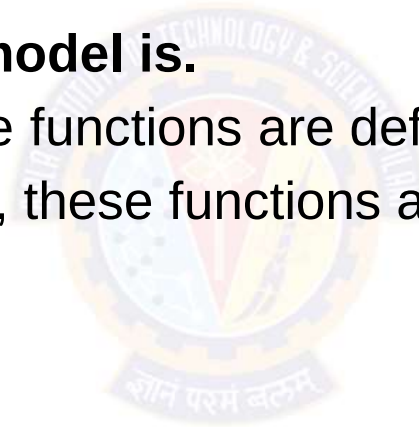


Fig. 1.1.1: Identify a wake word.

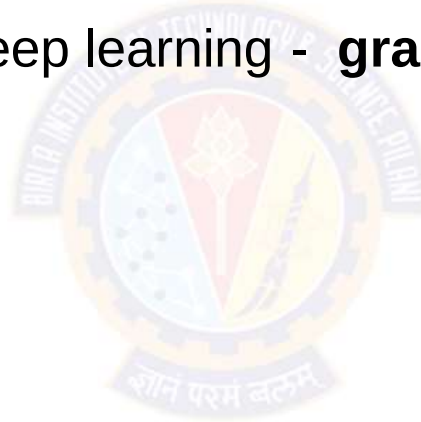
3. Objective Function

- Learning means improving at some task over time.
- **Objective functions are formal measures to denote how good (or bad) a mathematical model is.**
- By convention, objective functions are defined so that lower is better.
- Because lower is better, these functions are sometimes called **loss functions**.



4. Learning Algorithms

- **Learning Algorithm** is an algorithm capable of searching for the **best possible parameters** for minimizing the loss function.
- Popular algorithm for deep learning - **gradient descent**.



Example of the Framework

- We have to tell a computer explicitly how to map from inputs to outputs.
- We have to define the problem precisely, pinning down the exact nature of the inputs and outputs, and choosing an appropriate model family.
- Collect a huge dataset containing examples of audio and label those that do and that do not contain the wake word.

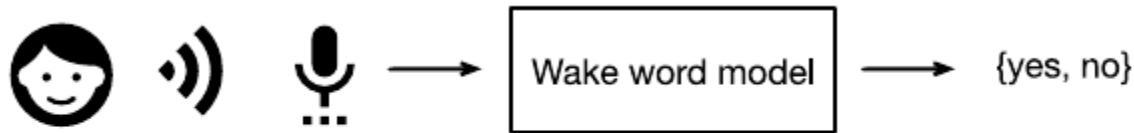


Fig. 1.1.1: Identify a wake word.

How to Create a Model?

- Start with a program whose behavior is determined by a number of parameters.
- How to determine the best possible set of parameters? Use data or examples.
- Parameters improve the performance of the program with respect to the loss function.

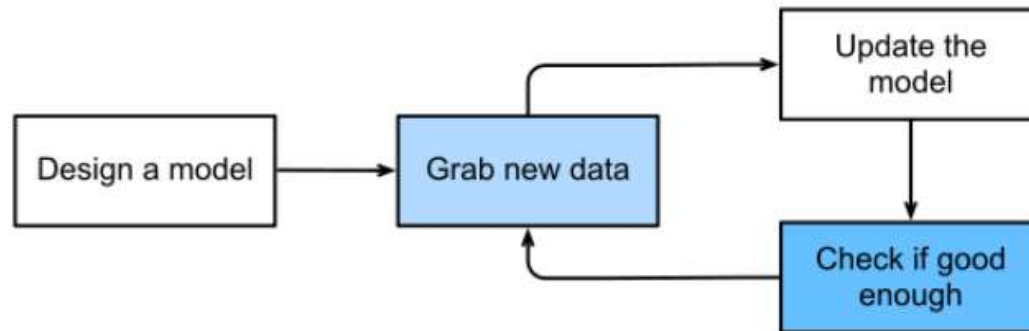


Fig. 1.1.2: A typical training process.

Train the Model

- Train the program with data to find the best set of parameters that improves the performance of the program with respect to the loss function.
- Once the parameters are finalised, we call the program a model.
 - Eg: The model receives a snippet of audio as input, and the model generates a selection among yes / no as output.
- The set of all distinct programs (input-output mappings) that we can produce just by manipulating the parameters is called a family of models.
 - Eg: Same model family should be suitable for "Alexa" recognition and "Hey Siri" recognition because they seem, intuitively, to be similar tasks.

Kinds of Deep Learning Problems



Learning Problems

1. Supervised Learning

- Training by examples
- The model is given a dataset which has the data and its labels
- Eg: Classification, Regression
- You will learn this in this course and Machine Learning Course

2. Unsupervised Learning

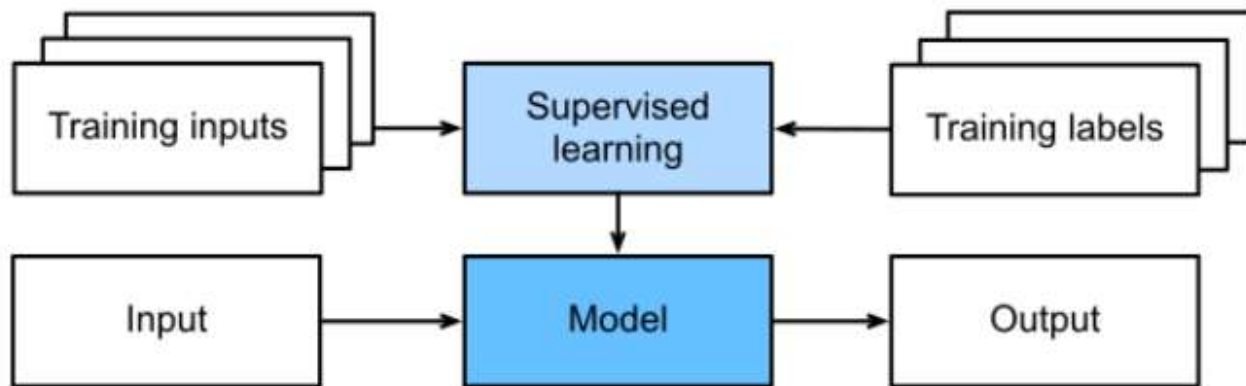
- Situations where we feed the model a giant dataset containing only the features
- Eg: Clustering, Generate synthetic data (uses GAN)
- You will learn this in Advanced Deep Learning Course and Machine Learning Course

3. Reinforcement Learning

- Develop an agent that interacts with an environment and takes actions over a series of time steps.
- You will learn this in Deep Reinforcement Learning Course

Supervised Learning

- Each **feature–label pair** is called an example.
- Goal is to produce **a model that maps any input to a label prediction**.
- The supervision comes into play because for choosing the parameters, we (the supervisors) provide the model with a dataset consisting of labeled examples, where each example is matched with conditional probability of a label given input features.



Supervised Learning Tasks

→ Regression

- ◆ how much or how many question

→ Classification

- ◆ Binary classification
- ◆ Multi-class classification
- ◆ Multi-label classification

→ Recommender systems

→ Sequence Learning

- ◆ Tagging and Parsing
- ◆ Automatic speech recognition
- ◆ Text to speech
- ◆ Machine translation



Further Reading
Dive into Deep Learning (T1)
Chapter 1



Thank You!